

Decarbonize industrial processes with the Turbo Heater

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Our innovative Turbo Heater technology is specifically designed to deliver high-temperature process heat at industrial scale without combustion of fossil fuels. This is achieved through a unique method that utilizes supersonic gas dynamics and shock waves to produce direct heating of gas mixtures. A novel approach to the electrification of high-grade process heat is introduced, decoupled from the physical limitations of heat transfer in conventional methods.

The Turbo Heater combines these features with the outstanding energy density of axial turbomachinery, and the high temperature capability of using materials already well known and proven in service in Siemens Energy's gas turbine products.

The result is a solution which is ready to scale and highly relevant to address the challenge of many hard-to-decarbonize industries.

Key Benefits

GHG Emissions: Rotating machinery can utilize clean energy (e.g. electric motor drive).

Direct Heating: Eliminate losses, fouling and leaks in heat exchangers for efficient (> 95%) heating of process flow.

Process Optimization:

- Very uniform temperature distribution (no hot spots) and precise temperature control
- Provide heat and flow in a single device, without separate circulation blower
- No combustion products added
- Near-instantaneous fluid heating simplifies downstream process control
- Can be implemented in hybrid configurations with fuel-fired backup

Size: Compact, energy-dense turbomachinery, heat input not constrained by exchanger surface area.

Scale-up: Technology inherently capable of high throughput, 2x flow << 2x size.

Inherently capable of **large-megawatt scale** and high temperature (1000 °C)

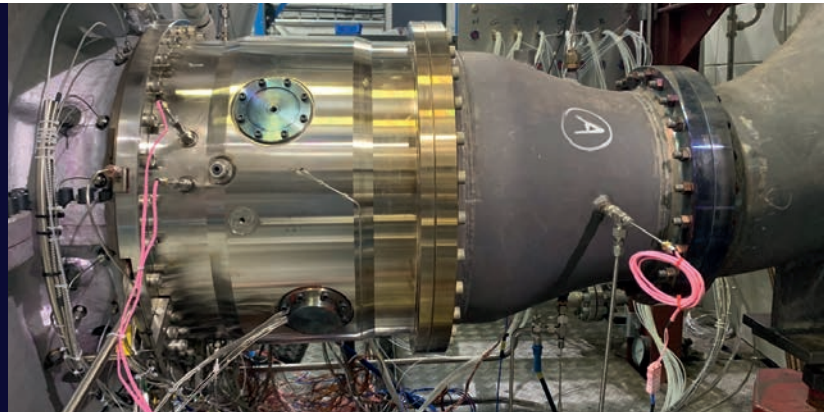
Direct heating of gas mixtures via fluid momentum (shock wave)

Eliminate fuel burning and decouple from **rate-limited** heat transfer

Potential use in combination with existing fuel-fired systems for **progressive phase-out**

Incremental technology maturation

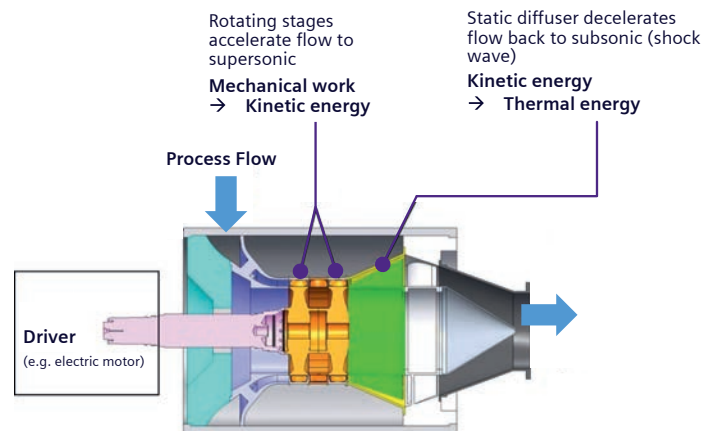
- Successful 1.6 MW demo exceeding 700 °C
- Higher temperature in development (> 1000 °C)
- Inherently scalable to double-digit megawatt and higher
- Built with materials and know-how already proven in Siemens Energy's Compression and Gas Turbine products



Decarbonize energy-intensive industrial processes via novel-class Turbomachinery



How the Turbo Heater works



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* In a parallel development, Siemens Energy has partnered with Technip Energies to jointly develop the Rotating Olefins Cracker (ROC), which applies the principle of shock wave heating turbomachinery to the steam cracking process to produce olefins such as ethylene.